

Training and validation curves

Project: Image-pattern Network Investigation

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Learning rate experiments

Without opening the old lesson, explain one key idea from Learning rate experiments. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Training a small network

Without opening the old lesson, explain one key idea from Training a small network. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Slopes and local change

Without opening the old lesson, explain one key idea from Slopes and local change. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Use the declared rule for Training and validation curves. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Training and validation curves. Proposed explanation: “The last checkpoint is always best”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Evaluate new checkpoints independently

Use $h = \text{ReLU}(x)$, $y = vh$, candidate v values 1, 1.5 and 2. Training pairs are (1, 2), (2, 4); validation pairs are (1, 1), (2, 2). Calculate both MSE columns and select by minimum validation MSE. Explain the disagreement with selection by training loss.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. W1 Recover a plotted point

At $v = 1$, compute both validation predictions, residuals and MSE on (3, 4.5) and (4, 6).

My reasoning and evidence

6. W2 Use the declared rule

Observed validation MSE at checkpoints 0 to 4 is [5, 2, 1, 1, 1.2]. Lowest loss wins with earlier exact ties. Which checkpoint is selected? What if training loss is lowest at checkpoint 4?

My reasoning and evidence

7. W3 Repair incompatible measures

A training report shows sum of squared errors 12 over six rows. Validation shows mean squared error 3 over two rows in the same target units. A learner says training error is four times larger. Repair the comparison.

My reasoning and evidence

8. W4 Evaluate new checkpoints independently

Use $h = \text{ReLU}(x)$, $y = vh$, candidate v values 1, 1.5 and 2. Training pairs are (1, 2), (2, 4); validation pairs are (1, 1), (2, 2). Calculate both MSE columns and select by minimum validation MSE. Explain the disagreement with selection by training loss.

My reasoning and evidence

9. W5 Limit a curve diagnosis

Training loss falls while validation loss rises. Give two checks before declaring model complexity the cause, and explain why repeatedly selecting with validation makes it development evidence.

My reasoning and evidence

10. Project checkpoint and reflection

Add a labelled training and validation curves evidence sheet to Image-pattern Network Investigation. Show how you can compute comparable training and validation losses at saved checkpoints. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

11. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

12. Pictorial case practice: Training and validation curves

Plot training and validation loss for checkpoints before opening final cases.

Supplied checkpoint losses: epoch 1 train4, val3; epoch 2 train2, val2; epoch 3 train1, val2.5. Selection: smallest validation loss.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

Checkpoint losses

	training	validation
epoch 1	4	3
epoch 2	2	2
epoch 3	1	2.5

Read a cell using both its row and column labels.

Training and validation curves

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Read training curve

Training loss falls 4 → 2 → 1.



2. Read validation curve

Validation loss falls 3 → 2 then rises to 2.5.



3. Choose checkpoint

Select epoch 2, not epoch 3, under the declared validation criterion.

CHANGE ONE THING

Epoch 3 validation loss is instead 1.5.

Explain the changed result using the matching lesson.

Can the original epoch 3 training loss 1 justify claiming best generalisation? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For your Image-pattern Network Investigation, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

13. Project portfolio milestone

Project: Image-pattern Network Investigation

Current milestone: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

For your Image-pattern Network Investigation, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

14. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Plot training and validation loss for checkpoints before opening final cases.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Select epoch 2, not epoch 3, under the declared validation criterion.

1. Read training curve
2. Choose checkpoint
3. Read validation curve

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Choose checkpoint Select epoch 2, not epoch 3, under the declared validation criterion. The training curve measures fit to examples used for learning. The validation curve measures a separate set used to choose a checkpoint. The final training loss is smallest here, but its validation loss has risen. Under the supplied selection rule, keep checkpoint two. A later final test must remain outside both learning and checkpoint selection.

Practice 2: explain the deciding evidence

Changed-case practice: Epoch3 validation loss is instead 1.5. Which conclusion is supported by the supplied facts?

1. Select epoch 3 under the unchanged criterion.
2. This one example guarantees correct results for every future input.
3. We can decide the result without examining the changed input or the procedure.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: Select epoch 3 under the unchanged criterion. Its measured validation loss is now below epoch 2's2; the training curve alone never settled selection.

Repair a missing building block

Review Learning rate experiments (ai-5/14). Goal: Run a controlled comparison of learning rates. Short practice: What should a fair rate comparison hold fixed? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Review Training a small network (ai-5/12). Goal: Trace repeated forward loss gradient and update steps. Short practice: Which parameter changed, and which part was not trained? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Review Slopes and local change (ai-5/9). Goal: Calculate an average rate of change over a stated interval. Short practice: Which calculation disproves “any increase helps”? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual output or error, change one input, then run again. Keep code, predictions, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
